

**DEPARTMENT OF CHEMICAL & BIOMOLECULAR ENGINEERING  
NORTH CAROLINA STATE UNIVERSITY**

**CHE 450**

**Homework Set 10**

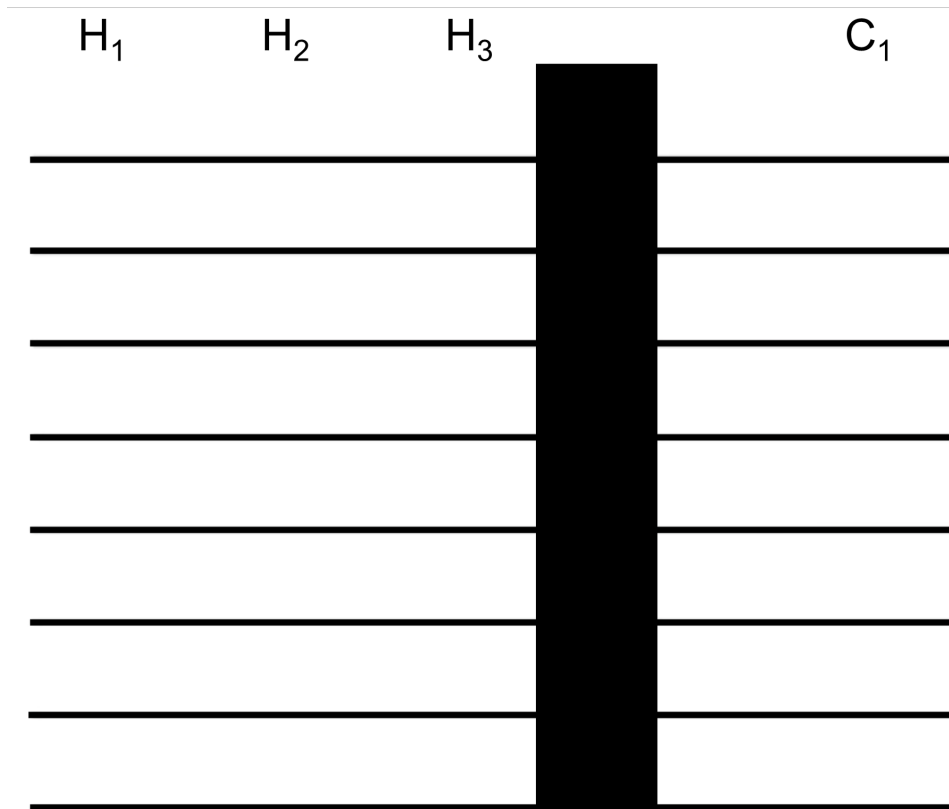
**Problem 1 (50%)** Use the algorithm provided in the lecture to complete the following Pinch Analysis for a heat exchanger network design. Assume a minimum approach temperature of 10°C.  
**(a, 30%)** Determine the pinch point (in °C) for the process streams shown in the table below.  
**(b, 20%)** Determine the hot and cold utility requirements in kW (remember, 1 kW = 1 kJ/s).

| Stream         | Actual Temperature |                | Effective Temperature* |                | $m \cdot C_p$ (kJ/s*°C) |
|----------------|--------------------|----------------|------------------------|----------------|-------------------------|
|                | $T_{in}$ (°C)      | $T_{out}$ (°C) | $T_{in}$ (°C)          | $T_{out}$ (°C) |                         |
| H <sub>1</sub> | 360                | 210            | 350                    | 200            | 5                       |
| H <sub>2</sub> | 260                | 60             | 250                    | 50             | 1                       |
| H <sub>3</sub> | 310                | 160            | 300                    | 150            | 1                       |
| C <sub>1</sub> | 40                 | 350            | 40                     | 350            | 4                       |

Minimum Approach = 10°C

\*Effective Temperature is the hot stream actual temperature minus the approach temperature.

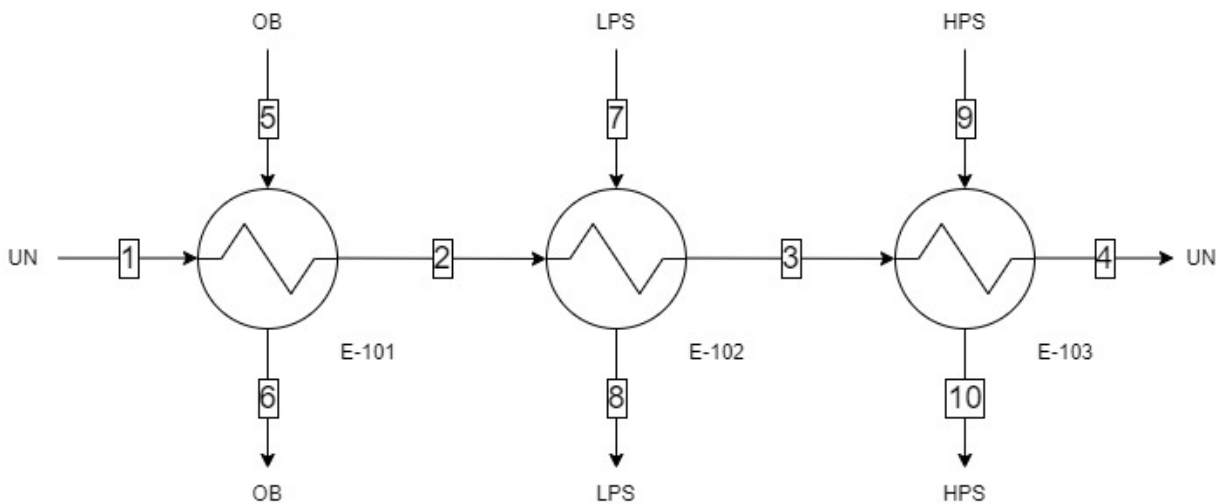
Below is a blank temperature interval diagram for use with this problem:



**Problem 2 (50%)** Examine the following network of three heat exchangers which is being used to heat the unobtanium (UN) feedstock using a combination of LPS, obtanium (OB, a recycle stream) and HPS. A stream table and equipment summary table appear on the following page. Your manager, Karen, is interested in saving energy. Assume that energy costs \$4/million BTU (million BTU is referred to as MMBTU). The fact that unobtanium *and* obtanium have the exact same physical properties as water remains a mystery to the entire corporation.

Your helpful, and familiar-seeming, coworker Cooper leans in and reminds you:

*In case you forgot,  $Q = U \cdot A \cdot LMTD$ , where  $A$  is the heat transfer area and  $LMTD$  is the log-mean temperature difference. Also recall that  $Q = \dot{m} \cdot C_p \cdot \Delta T$ .*



- (a, 15%) Which heat exchanger has the highest U-value in BTU/(°F·ft<sup>2</sup>)? What does this imply?
- (b, 15%) Which heat exchanger has the lowest U-value in BTU/(°F·ft<sup>2</sup>)? What does this imply?
- (c, 5%) Which heat exchanger would you recommend cleaning? Why?
- (d, 10%) Assume E-103 has now been cleaned and it has a U-value of 200 BTU/(°F·ft<sup>2</sup>). What is the clean duty ( $Q_{\text{clean}}$ ), assuming the same LMTD?
- (e, 5%) It costs \$30,000 to clean one heat exchanger. What is the benefit-to-cost ratio (BCR) for cleaning E-103? Assume a one-year timeframe for the benefits.

**Table 1. Stream table accompanying Problem 2.**

|                | Units |              |              |              |              |              |              |             |             |             |             |
|----------------|-------|--------------|--------------|--------------|--------------|--------------|--------------|-------------|-------------|-------------|-------------|
| Stream Number  |       | 1            | 2            | 3            | 4            | 5            | 6            | 7           | 8           | 9           | 10          |
| From           |       |              | E-101        | E-102        | E-103        |              | E-101        |             | E-102       |             | E-103       |
| To             |       | E-101        | E-102        | E-103        |              | E-101        |              | E-102       |             | E-103       |             |
| Phase          |       | Liquid Phase | Liquid Phase | Liquid Phase | Liquid Phase | Liquid Phase | Liquid Phase | Vapor Phase | Vapor Phase | Vapor Phase | Vapor Phase |
| Temperature    | F     | 65.0         | 126.2        | 348.8        | 568.5        | 212.0        | 151.2        | 475.0       | 373.8       | 700.0       | 593.5       |
| Pressure       | psia  | 1450.4       | 1450.4       | 1450.4       | 1450.4       | 1450.4       | 1450.4       | 145.0       | 145.0       | 652.7       | 652.7       |
| Mass Flows     | lb/hr | 440924.5     | 440924.5     | 440924.5     | 440924.5     | 440924.5     | 440924.5     | 2204622.6   | 2204622.6   | 2204622.6   | 2204622.6   |
| WATER          | lb/hr | 440924.5     | 440924.5     | 440924.5     | 440924.5     | 440924.5     | 440924.5     | 2204622.6   | 2204622.6   | 2204622.6   | 2204622.6   |
| Mass Fractions |       |              |              |              |              |              |              |             |             |             |             |
| WATER          |       | 1.0          | 1.0          | 1.0          | 1.0          | 1.0          | 1.0          | 1.0         | 1.0         | 1.0         | 1.0         |

**Table 2. Equipment summary table accompanying Problem 2.**

| Name                               | E-101      | E-102       | E-103       |
|------------------------------------|------------|-------------|-------------|
| Minimum temperature approach [F]   | 25.0       | 25.0        | 25.0        |
| Inlet hot stream temperature [F]   | 212.0      | 475.0       | 700.0       |
| Inlet hot stream vapor fraction    | 0.0        | 1.0         | 1.0         |
| Outlet hot stream temperature [F]  | 151.2      | 373.8       | 593.5       |
| Outlet hot stream vapor fraction   | 0.0        | 1.0         | 1.0         |
| Inlet cold stream temperature [F]  | 65.0       | 126.2       | 348.8       |
| Inlet cold stream vapor fraction   | 0.0        | 0.0         | 0.0         |
| Outlet cold stream temperature [F] | 126.2      | 348.8       | 568.5       |
| Outlet cold stream vapor fraction  | 0.0        | 0.0         | 0.0         |
| Heat duty [Btu/hr]                 | 29095991.7 | 108747171.0 | 129810342.0 |
| Actual exchanger area [sqft]       | 2255.4     | 5031.4      | 7122.5      |
| Average U (Dirty) [Btu/hr-sqft-R]  |            |             |             |
| Average U (Clean)                  |            |             | 200.0       |